Steam Game Analytics & Ranking System

A PROJECT REPORT

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Submitted by

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In partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE & ENGINEERING

Of

FACULTY OF ENGINEERING AND TECHNOLOGY

Delhi NCR Campus, Modinagar, Ghaziabad (U.P.)

MAY 2023

SRM INSTITUTE OF SCIENCE &TECHNOLOGY

(Under Section 3 of UGC Act, 1956)

BONAFIDE CERTIFICATE

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ABSTRACT

Malaria is a major burden on our health-care system, and it is the leading cause of mortality in many developing countries. It is endemic in specific parts of the world, which means that the illness is always present in such locations. As a result, early testing is critical for detecting malaria and saving lives. As a result, it inspires us to make the malaria conclusion more appealing and faster. To overcome this issue, specialist technology is required. The RDT and microscopic diagnosis are the most often used clinical techniques for malaria diagnosis. RDT is a gadget that is both powerful and speedy.

It also does not necessitate the presence of a qualified clinical specialist. However, when compared to a light magnifying device, RDT has a number of drawbacks, including being susceptible to heat and muck, as well as being more expensive. Although microscopic diagnosis analysis frameworks are free of these flaws, they do necessitate the presence of a trained microscopist. In the therapeutic field, Machine Learning has recently emerged as truly newsworthy. Machine Learning techniques have proven to be effective in bringing an illness to a close. The main goal of this malaria identification framework is to address the shortcomings of the current system by automating the malaria detection process using Machine Learning and Image Processing.

Keywords: RDT, Image Processing

ACKNOWLEDGEMENTS

We would like to thank our guide, Mr. Arvind Kumar, for her valuable guidance, constant encouragement, personal caring, timely aid and offering timely assistance as well as providing us a wonderful environment in which to explore ourselves. Throughout the project, she has extended bright and agreeable assistance to us, despite her hectic schedule.

We would also want to express our gratitude to Dr. Naveen Ahlawat, Professor and Head, Department of B.Tech First Year, SRM Institute of Science and Technology (Delhi-NCR Campus), for his helpful recommendations and support during the project's duration. We would like to express our gratitude to the personnel and students at the SRM Institute of Science and Technology's Computer Science and Engineering Department for their assistance during our investigation. Finally, we want to express our gratitude to our parents, family, and friends for their unwavering love, constant assistance, and unwavering support.

Saurabh Gupta

Akash Goyal

Tenjeet Kumar

Rahul Garg

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ABBREVIATIONS

CNN	Convolution Neural Network
VGG	Visual Geometry Group

CHAPTER 1
INTRODUCTION

1.1. The Idea behind It

Malaria is a mosquito-borne disease caused by the plasmodium parasite, which is transmitted through mosquito bites. Every year, an estimated 300-500 million people develop malaria over the world, resulting in 1.5-2.7 million deaths [2]. According to the World Health Organization (WHO), around 219 million cases of malaria were identified in 2017, resulting in 435,000 deaths worldwide [3]. Malaria is a fatal infection that is more frequently detected in rural areas where clinical decision-making and medical service options are limited. Professionals use the widely accepted light microscopy method to determine the presence of malaria. Regular light microscopy for malaria detection examines both good and bad stained blood spreads. In areas where malaria is a serious problem, microscopy is used extensively. The reliance on gifted specialists, of which there is a basic deficiency, is a significant downside of this technique. In Ghana, a cross-country study discovered 1.72 magnifying lenses per 100,000 people, but only.85 trained research centre employees per 100,000 people [4]. This leads to a postponement and an incorrect conclusion. Findings are frequently made only on the basis of clinical indicators and side effects, which is prone to error, and this erroneous analysis leads to higher mortality, drug blockage, and the financial burden of acquiring unnecessary prescriptions.

By preventing major malaria cases, early and precise malaria diagnosis and treatment can help a patient and save many lives. From one end of the globe to the other, 5 billion people are still without access to malaria prevention and treatment. As a result, there is a necessity for a strong elective that will assist in providing access to top-notch decision-making on a regular basis that is now unavailable. This investigation focuses on developing a precise malaria detection model that can be implemented without relying on skilled specialists, as well as assessing the model's precision to achieve top-notch

results. Automated image analysis programming could overcome the most significant limitation of the generally accepted microscopy technique, which is the reliance on human experts for analytic precision of the results. Computerizing the recognition cycle entails combining data with common approaches and executing them to provide rapid and effective results.

1.2. The Plan

Machine Learning is a branch of science in which decisions are made by analysing and analysing data in order to achieve the best result. In the last ten years, the health-care industry has seen tremendous growth, with a lot of research being done to computerise medical services and make interaction more available to everyone. Many studies have been conducted on various Machine Learning models in order to automate the conclusion interaction. Machine Learning models necessitate a great deal of fine tweaking, factor analysis, and component design. With more information, the Machine Learning approach isn't as adaptable. AI necessitates designing and element preparation, which is far from a simple process. Profound learning models are reliable and adaptable, with a greater accuracy rate. Convolutional Neural Networks (CNN) have proven to be extremely effective in a variety of computer vision tasks [4]. Convolutional Neural Networks convert the volume of an input picture into an output volume with a class mark. Include extraction and element design are not required in Convolutional displaying, as extricating the highlights with the most variation, such as which component is more important, is sufficient. Pictures are accepted as input by CNN models, and the image is transferred through multiple layers of the CNN model, where it is analysed in new regions. As a result, Convolutional neural Organization is an excellent way for identifying Malaria from patient platelet images.

The RDT and miniscule finding are the most often used clinical procedures for Malaria detection. RDT is a practical and time-saving device. It also does not necessitate the presence of a trained clinical specialist. However, as compared to a light magnifying

lens, RDT has a few drawbacks, such as being vulnerable to heat and wetness and being more expensive. The shortcomings have no influence on the minute analysis frameworks, but they do necessitate the presence of a trained microscopist. In the healthcare field, AI has recently emerged as truly newsworthy. In the examination of a disease, AI techniques have proven to be quite successful.

CHAPTER 2

Literature survey

2.1 Malaria and Its Diagnosis

Malaria is a blood infection caused by the Plasmodium parasite, which is transmitted from one person to another by a specific type of mosquito. Malaria is one of the world's worst diseases, and it continues to be one of the world's leading killers of children [2]. Malaria sickness can cause paleness, hypoglycemia, or cerebral malaria, in which blood arteries cannot transport blood to the brain due to a delay in diagnosis and treatment. Cerebral malaria can result in a trance-like state, learning disabilities, and death [5]. Malaria may be detected early, which reduces mortality. The most renowned malaria determination technique is the widely accepted infinitesimal conclusion, in which malaria is studied minutely by colouring good and poor blood films on a glass slide to visualise malaria parasites. Master specialists imagine the pre-arranged slides beneath a magnifying lens, and this technique's widespread acceptance can be attributed to its simplicity and ability to discern the presence of parasites.

Nonetheless, the cycle is labor-intensive and time-consuming, and it necessitates master and well-prepared medical care workers. The little study is wasteful because to a lack of and inaccessibility of skilled specialists and health-care resources in remote country regions. Because of adverse effects that lead to a false diagnosis and unsuitable therapy, the vast majority of people in rural areas use some form of self-treatment. Lab misdiagnosis is also totally predicted in endemic countries like India and South Africa, according to studies [6]. Malaria misdiagnosis can be caused by two major factors: enormous responsibility and limited human resources, with the most important explanation being a shortage of qualified specialists. Around 80% of the global malaria problem was carried by fifteen countries in Sub-Saharan Africa and India [3]. There is a definite need for talented specialists in countries with high malaria rates; according to a research conducted in Ghana, there are only two experts for every 100,000 cases,

resulting in erroneous readings due to the professional's weight [7]. In countries where malaria isn't as common, there aren't as many skilled professionals; experts aren't as well-prepared as they need to monitor malaria cases on a regular basis. The location cycle will be computerised, allowing for simple and powerful determination from one side of the world to the other. Information can be acquired from all around the world, and a model can be built based on that data to produce more precise results.

The optimum quality level strategy for malaria conclusion on Giemsastained slender and thick blood films is light microscopy, which needs a high level of ability. Because this isn't always available in certain locations, Fast Symptomatic Tests (RDTs) are increasingly being used. Despite this, RDTs have limitations, and the presence of tainting species is not always certain when a malaria diagnosis is made. The master's centre obsessive comprehension is the most important approach for malaria analysis microscopy and its assessment precision. In essence, such a system contains a humanistic error in terms of subjectivity, resulting in contradicting as well as less analytic precision. Fostering a PC-assisted malaria parasite finding device has gained relevance in today's obsessive administrations, where a clinician will be assisted in making a superior option toward malaria conclusion, in order to increase analytic accuracy by minimising such subjectivity. AI procedures have substantial commitments in today's indicative framework enhancement for achieving higher demonstrative accuracy in clinical imaging informatics like microscope, ultrasonic imaging, X-ray, and CT [8]. The most important and beneficial tool for a neurotic assessment of various infections, haematological upheaval, oral malignant development, bosom sickness, cervical disease, and so on is infinitesimal picture examination. Fringe blood smear screening, like other minute picture examinations, is one of the most basic demonstrative ways for detecting haematological difficulties (sickness, thalassemia, and so on) and parasitic disease (malaria, filaria) in the blood. Pathologists frequently use a light magnifying lens to recognise disease in erythrocytes due to the variety and structural alterations of the erthrocytes when diagnosing malaria [9].

2.2 Machine Learning Applications in Health Care

PC-assisted symptomatic gadgets that use AI (ML) algorithms on microscopic blood smear images can potentially reduce clinical weight by assisting with emergency and sickness translation. These devices cycle clinical images for standard appearances and include obsessive aspects to improve clinical independence. As a result, picture-based clinical finding and risk assessment have improved significantly with PC-assisted analytic devices. Transfer learning, support vector machines, computed relapse, and k-closest neighbour computation all perform well in this situation, according to Wei Chen F.Z.'s research. Pre-processing tasks such as stain removal from fringe blood smear images and drive sound reduction were also completed. Mustafa et al presented thresholding as a pre-processing phase in their research, and it is considered one of the most important pre-processing procedures in this errand. The Fluffy C-Mean computation, Wolf's technique, Bradley's strategy, and Bernsen's strategy have all been examined in their work. Each of these methods looks into a different angle when it comes to the malaria image dataset. However, the majority of these devices used to detect malaria use hand-crafted element extraction computations 10 that are enhanced for specific datasets and prepared for explicit variations in source hardware, aspect, position, and direction of the area of interest (return on investment) [8]. Many researchers have focused their efforts on malaria detection and have achieved remarkable results, but as technology advances and more information becomes available, a new discipline of computerised reasoning known as Profound Learning is gaining popularity. Profound learning is a man-made neuralpower task that mimics the human cerebrum's operations in managing information and designing designs for independent application. Profound learning is a subtype of AI in man-made reasoning (simulated intelligence) that uses networks to learn from unstructured or unlabeled data without assistance. It's also known as profound neural network or profound neural learning. Analysts have discovered new techniques to make data analysis and forecasting more productive and precise than AI models, thanks to advancements in

technology. Previously, scientists used SVM (Backing Vector Machines) to analyse the Malaria dataset, which required them to preprocess the data to remove commotions and protect the important highlights, as well as improve picture contrast and light accuracy. Division, highlight calculation and selection, and then characterisation were the next stages of the interaction. After much research and testing, the AI model was not as precise as it appeared to be expected to be in order to replace manual evaluation. Further research into deep convolutional networks yielded amazing results in a variety of image datasets. The Alex Net neural network, which won honours in the ImageNet challenge, performed exceptionally well on the picture recognition dataset.

2.3 Deep Learning Applications in Health Care

By self-finding the traits from 11 basic pixel information and performing start to finish include extraction and arrangement, data driven deep learning (DL) techniques have supplanted the presentation of human tailored highlight extraction instruments. Convolutional neural organisations (CNN), a type of DL model, have demonstrated promising results in image arrangement, recognition, and confinement assignments. The availability of vast amounts of clear information is credited with CNNs' promising performance. Move it is embraced to learn systems in settings of restricted information accessibility, such as clinical images. Var et al suggested PC-assisted analysis strategies for recognising malaria parasites using pre-prepared convolutional neural networks as element extractors. The CNN models are pre-trained on large datasets like ImageNet in order to apply the information learnt as traditional picture highlights to the objective task. The pre-trained loads serve as a good introduction, and they are found to perform better than preparing the model with haphazardly instated loads. Writing investigations reveal the application of common ML and information-driven DL algorithms to the detection of malaria parasites in thin blood smear images. Dong et al investigated the use of portion-based calculations, such as support vector machines (SVM) and convolutional neural networks (CNNs), to arrange tainted and regular cells. A small number of sectioned red platelets (RBCs) were randomly divided into train, approval,

and test sets. The CNNs were found to have an arrangement exactness of more than 95%, outperforming the SVM classifier, which had a precision of 92 percent. The CNNs self-discovered the highlights from the raw pixel data, necessitating just minor human intervention in the robotized finding process. He used cross-approval learns at the cell level to evaluate the performance of custom and pre-prepared CNN models in grouping parasitized and normal cell images. The custom CNN outperformed the pretrained Alex Net model with a precision of 97.37 percent, according to preliminary results.

CHAPTER 3

SYSTEM ANALYSIS

This investigation makes use of deep learning. We're making full use of these techniques' capabilities. Deep Learning is fascinating, and we believe that this fascination is justified, as Deep Learning has already demonstrated its worth by dramatically transforming PC vision. Deep Learning has progressed to the point where it can now energise image grouping and PC vision. Any deep learning model requires a lot of data, so we're using a variety of data enhancement tactics in addition to the large number of photos we've collected.

3.1. Transfer Learning

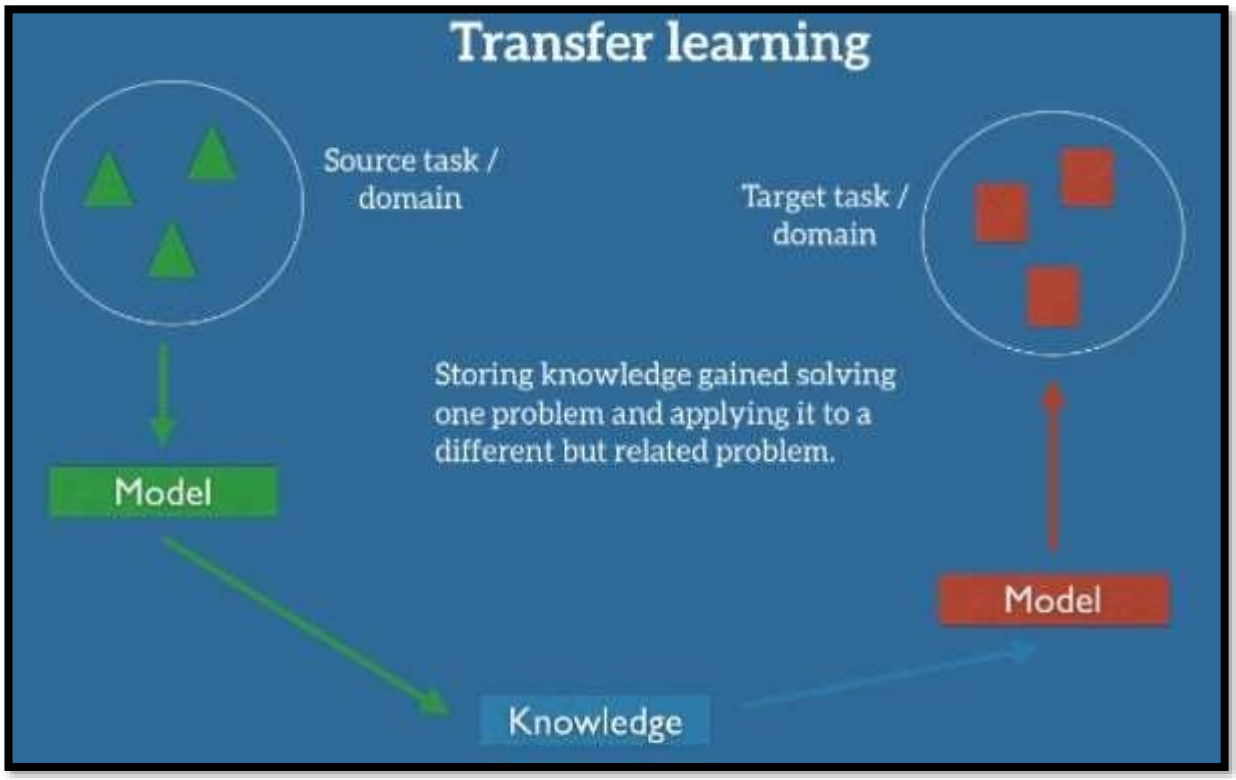

Figure 3.1: Transfer Learning

If we opt to construct a model for any undertaking and space A, we assume that we will be offered marked information for a similar task and area in the Transfer learning

situation of Machine Learning. Propose, for now, that our model has a mission, such as perceiving objects in photographs, and that our discoveries emanate from a space, such as photographs shot in San Francisco coffeehouses.

We can now build a model based on this data and expect it to excel at extracting information from the mission and space that has never been seen before. We re-quire consigned information for the precise errand or place that we may utilise to develop a nascent model B so we can expect that it should excel on this information when we are supplied information for an alternate assignment or region B.The normal guided mastery strategy fizzles when we require more precise knowledge for the task or area for which we are preparing a predictable model. Using a model that has been developed on a large area, such as daytime photographs, we may prepare a model to perceive walkers on night time pictures. Nonetheless, we will see an optical reduction in legitimacy outcomes since the model has picked up on the difference in its preparation data and is cautious to sum up to the fledgling area. We couldn't employ an ongoing model to enlighten a model to direct an incipient work like identifying bikers since the markers between the undertakings are unique.

Move learning enables you to negotiate your way out of these situations by employing previously marked data from a similar task or place. We aim to apply the knowledge we gained from solving the source predicament to our main challenge in the source space.

3.2 Convolution Neural Network (CNN)

Convolutional Neural Networks are similar to traditional Neural Networks in that they are composed up of neurons with learnable loads and inclinations, just as traditional Neural Networks are. Any neuron stores a large amount of data, does a spot item, and then performs a non-linearity as needed. The entire endeavour, from the most basic picture pixels on one side to class scores on the other, always transmits a single unique

score work. They do indiscriminate work on the last (usually related) layer (for example, SVM/Softmax), and the tips/tricks we learned for thinking about unimpressive Neural Networks apply in general.

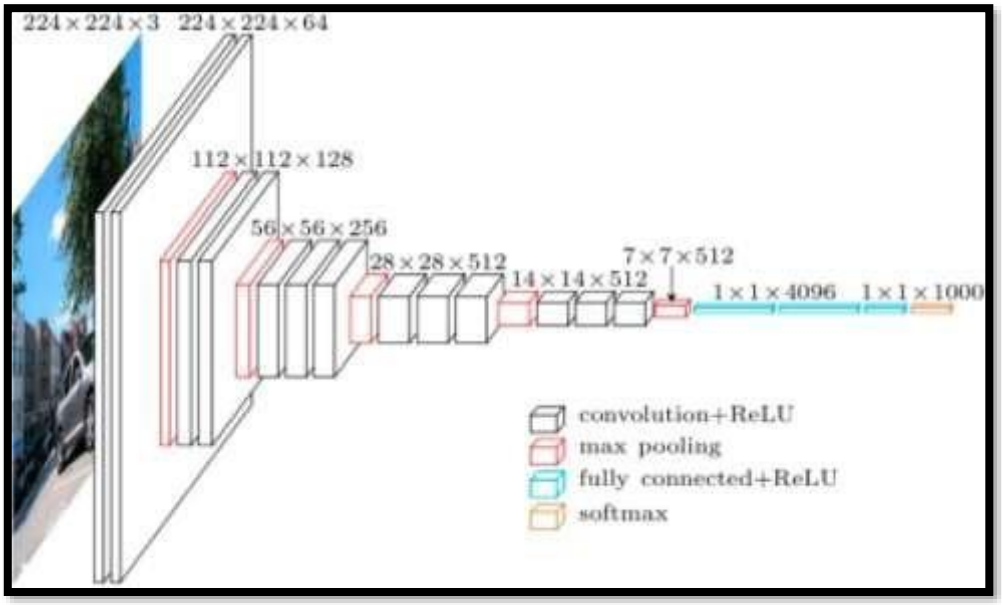

Figure 3.2: Convolution Neural Network

The translucent ConvNet components provide the verbal articulation that the information sources are pictures, allowing one to encode those features into the design. Similarly, forward work is more manageable to complete, and the amount of constraints in the association is substantially reduced. A core Convolutional Neural Network consists of a series of layers, each of which uses a differentiable limit to shift more than one proportion of sanctionments into another. Convolutional Neural Network structures are built utilising three types of layers: Convolutional Layer, Pooling Layer, and Plenarily Connected Layer (just like a regular Neural Network).

3.3. System Architecture

The application will gradually acquire the picture outlines. After that, information rehandling/image handling methods such as cleaning, increasing, and changing will be used to the obtained data. We're using our dataset to create a VGG neural network model with Keras, expecting test data, and then going over it for a consistent output.

CHAPTER 4

SYSTEM DESIGN

4.1 Data Description

The 'National Institutes of Health' provided the dataset for this project, which contains 30000 cell images.

This dataset is divided into two folders: the test and train envelopes, each with its own set of testing and preparation images. There are also two more organisers: a parasitized envelope containing malaria-tainted cell images and an uninfected envelope with normal cell images. We don't have to deal with difficulties of imbalanced information because we have a comparable appropriation of cell photos, thus there are fewer chances of it being one-sided towards a specific class.

4.2 Models

4.2.1 VGG

In their publication "Profoundly DEEP CONVOLUTIONAL NETWORKS FOR SIZABLY VOLUMINOUS-SCALE IMAGE APPERCEPTION," Andrew Zisserman and Karen Simonyan of Oxford University's VGG (Visual Geometry Group) Lab proposed VGG-16. This paradigm took first and second place in the aforementioned relegations in the 2014 ILSVRC challenge.

VGG Net has figured out how to abstract aspects that can disunite artefacts (highlight extractor), and it's now being utilised to agnize non-ostensible artefacts. The goal of VGG was to improve order precision by increasing CNN depth. For object perception, VGG 16 and VGG 19, which have 16 and 19 weight layers, respectively, were used. VGG Net trains a convolutional layer stack with a fine-tuned channel size of 33 and a phase of 1 using input from 224224 RGB pictures.

VGG Net has figured out how to distribute ascribes (including extractors) that can disassemble products, and it's now being used to torment non-ostensible artefacts. The goal of VGG was to increase the depth of CNNs in order to improve characterisation precision. For object perception, VGG 16 and VGG 19, which have 16 and 19 weight layers, respectively, were used. VGG Net trains a convolutional layer stack with a fine-tuned channel size of 33 and a phase of VGG Net by using information from 224224 RGB images. The VGG-19 is a 19-layer deep Convolutional Neural Network. VGG-19 is one of the VGG model's variants. VGG comes in a variety of forms, including VGG-11, VGG16, and others. Visual Calculation Gathering (VGG) was created at Oxford's and was given the term VGG as a result. It makes use of deep convolutional neural layers to improve precision. VGG19 Engineering has 16 convolutional layers and three fully associated layers, as well as five max-pool layers. We used move figuring out how to prepare our model on the ImageNet dataset in this investigation. It's a photo database with 14,197,122 images organised according to the WordNet pecking order. This is a push to assist picture and vision researchers, including specialists, understudies, and others. Cells with Stain Standardization used VGG-19 on the malaria dataset by using move learning with frozen layers, planned model without any preparation, and move learning with modifying the model. With each of the 19 teachable layers on the malaria dataset, the underlying model was planned without any prior preparation.

CHAPTER 5

SYSTEM REQUIREMENTS

5.1 Software Requirements

Operating System: Windows and macOS

Frameworks: Keras, TensorFlow 1.16, NumPy 1.18, SciPy 1.4.1, Tensor Board 1.15.0, TensorFlow Probability 0.8.0

Language: Python 3.8 and above

5.2 Hardware Requirements

CPU Type: Intel Pentium 4 and above

Clock Speed: 3.0 GHz and above

RAM Size: 8 GB and above

Hard Disk Capacity: 80 GB and above

5.3 Models

5.3.1 Keras

By allowing us to construct previously unseen libraries and frameworks, deep learning has revolutionised the artificial intelligence business. Deep Learning is a sub-discipline of Machine Learning that uses deep artificial neural networks to address Machine Learning problems.Keras is a python-based neural network library that typically uses Tensor Flow, Microsoft CNTK, or Theano as a backend high-level library. It offers sklearn-style (Python-based) Neural Network creation APIs. It enables users to design neural networks fast and easily without having to consider the technical complexity of tensor/vector algebra, computational methodologies, or optimization approaches.

Keras was created with the goal of enabling test-driven development and rapid prototyping. In order to do effective research, professors must travel from notion to conclusion in the shortest period possible. Every firm is attempting to include Deep Learning in some manner, and Keras provides an API that is fundamentally simple to use and understand, allowing you to test and develop Deep Learning applications with the least amount of work.

Keras is a high-level library with a Theano or TensorFlow runtime that is ready to use. It performs flawlessly on both the CPU and the GPU. Keras improves all neural network models proximally, including entirely connected, convolutional, pooling, recurrent, embedding, and so on. These models can also be combined to create more complex models. The modular architecture of Keras makes it very articulate, versatile, and well-suited to cutting-edge research. Keras is essentially a Python-based framework, which simplifies debugging and research.

5.3.2 Matplotlib

Python visualisation packages include NumPy, a numerical calculus extension for Python, and Matplotlib. It provides an object-oriented plotting API for GUI toolkits like wxPython, Tkinter, GTK, and Qt. A procedural "pylab" interface based on a state machine (equivalent to OpenGL) that is meant to appear like MATLAB is also available, but it is not recommended. Matplotlib is used by SciPy to plot results.

Matplotlib 2.0.x was added to Python versions 2.7 and 3.6, and Matplotlib 1.2 was added to Python 3. Matplotlib 1.4 is the most recent upgrade for Python 2.6. Matplotlib has signed the Python 3 Verbal Agreement, promising not to revive Python 2 until 2020.Pyplot is a Matplotlib application that mimics the MATLAB user interface. In any event, Matplotlib is designed to be as user-friendly as MATLAB, with the added bonus of being open-source and free.

5.3.3 Google Colab

As you might have gathered from the name, Google Colab is a Google product. It's essentially a free-to-use cloud-based notebook environment. It offers features that allow you to edit archives in the same way as Google Docs does. Colab supports a variety of normal and undeniable level AI libraries that may be conveniently placed in your note pad.

To put it another way, Google Colab is a completely cloud-based version of the Jupyter notebook environment. The most important feature of Google collab is that it doesn't require any configuration, and the notebooks you create can be changed in real time by your team members, similar to Google Docs papers. Colab supports a variety of machine learning libraries that may be easily loaded into your notebook.

5.3.4 Tkinter

Tkinter is Python's default Graphical User Interface library. When Python is combined with Tkinter, it provides a quick and easy way to create graphical user interfaces. Tkinter provides a well-organized article-based access point to the Tk GUI toolkit. Tkinter gives the Tk GUI toolkit a rich object-oriented interface.

5.4 Languages

5.4.1 Python

Python is an easy-to-learn programming language with a wide range of applications. Python includes a basic yet useful OOPS framework as well as sophisticated high-level data structures. Python's attractive syntax and dynamic scripting, together with its interpreted architecture, make it a good language for scripting and rapid programme development in a wide range of subjects on a wide range of systems.

Features of Python: -

- Simple
- Easy to learn
- Portable
- High-level language
- Object-Oriented
- Extensible
- Interpreted
- Free and open space
- Embeddable
- Extensive Libraries

CHAPTER 6

IMPLEMENTATION

6.1 Data Collection

To prepare, we used Google Open-Source Pictures and Kaggle to acquire informational indexes. A data set of over 30000 photos was created and divided into separate directories. To improve the efficiency of our model, we cleansed the data set for corrupt files. This dataset is divided into two folders: the test and train envelopes, each with its own set of testing and preparation images. There are also two more organisers: a parasitized envelope containing malaria-tainted cell images and an uninfected envelope with normal cell images. We don't have to deal with difficulties of imbalanced information because we have a comparable appropriation of cell photos, thus there are fewer chances of it being one-sided towards a specific class.

6.2 Training

We'll use VGG-19 for training, then transfer learning to add some of our own layers to the bottom, as well as adjusting the input layer.

```
r = model.fit(
  training_set,
  validation_data=test_set,
  epochs=5,
  steps_per_epoch=len(training_set),
  validation_steps=len(test_set)
)

Epoch 1/5
690/690 [==============================] - 385s 558ms/step - loss: 0.2417 - accuracy: 0.9156 - val_loss: 0.2846 - val_accuracy: 0.8933
Epoch 2/5
690/690 [==============================] - 377s 546ms/step - loss: 0.2339 - accuracy: 0.9182 - val_loss: 0.3215 - val_accuracy: 0.8895
Epoch 3/5
690/690 [==============================] - 377s 546ms/step - loss: 0.2434 - accuracy: 0.9164 - val_loss: 0.2680 - val_accuracy: 0.8989
Epoch 4/5
690/690 [==============================] - 384s 557ms/step - loss: 0.2319 - accuracy: 0.9199 - val_loss: 0.4053 - val_accuracy: 0.8628
Epoch 5/5
690/690 [==============================] - 384s 556ms/step - loss: 0.2208 - accuracy: 0.9231 - val_loss: 0.2878 - val_accuracy: 0.8988
```

Figure 6.1: Training snippet

We're training our model in the image above. And we're running 5 epochs, and as you can see from the graph, we're achieving around 92 percent accuracy.

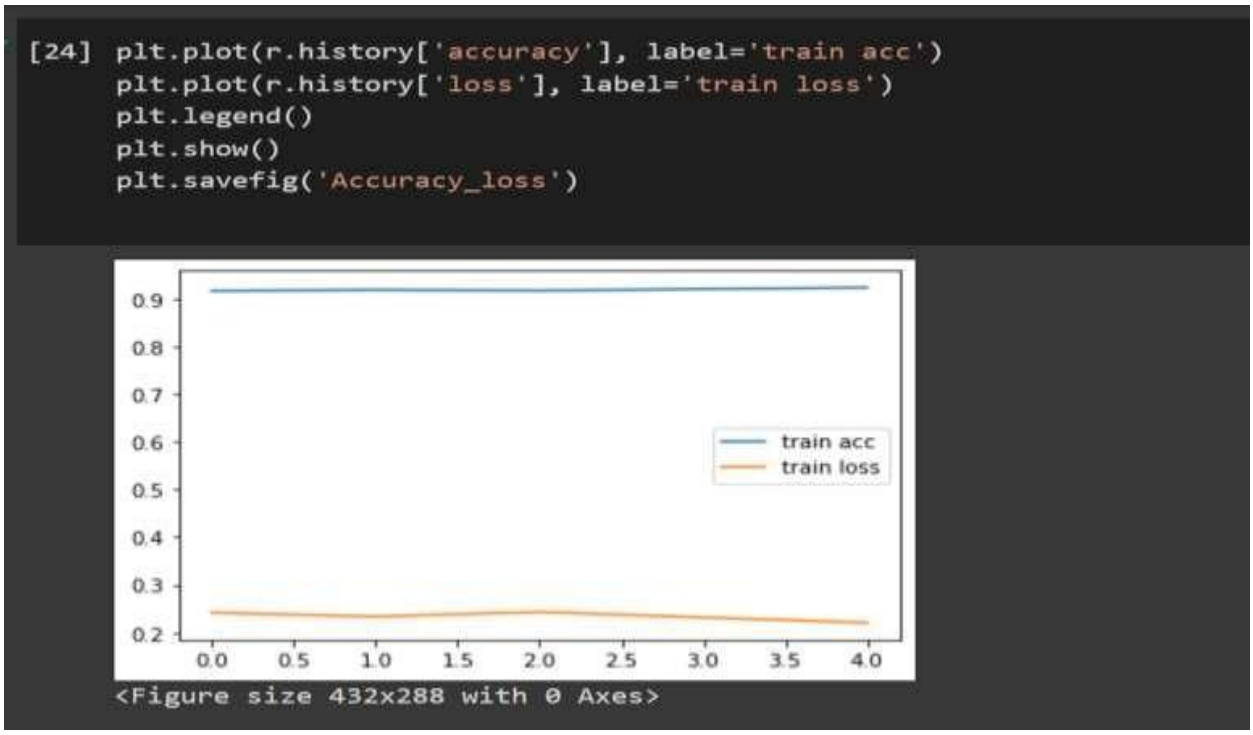

Figure 6.2: Training Accuracy-loss graph

The graph plot between training accuracy and trained loss is shown in the image above. Train loss is represented on the X-axis, while train precision is represented on the Y-axis.

6.3 Working

The input image is pre-processed to remove any unwanted clam or from the RGB cell image. The pre-handled image is then used to plot pictures and names in order to understand how a tainted cell and an uninfected cell appear to be. Make a model of a Convolutional Neural Network. We next prepared the Train Information and anticipated it on the Test Information at that time. The next stage is the organisation stage, in which the information is represented by element vectors, with the outcome being an ordered term such as parasitic or non-parasitic.

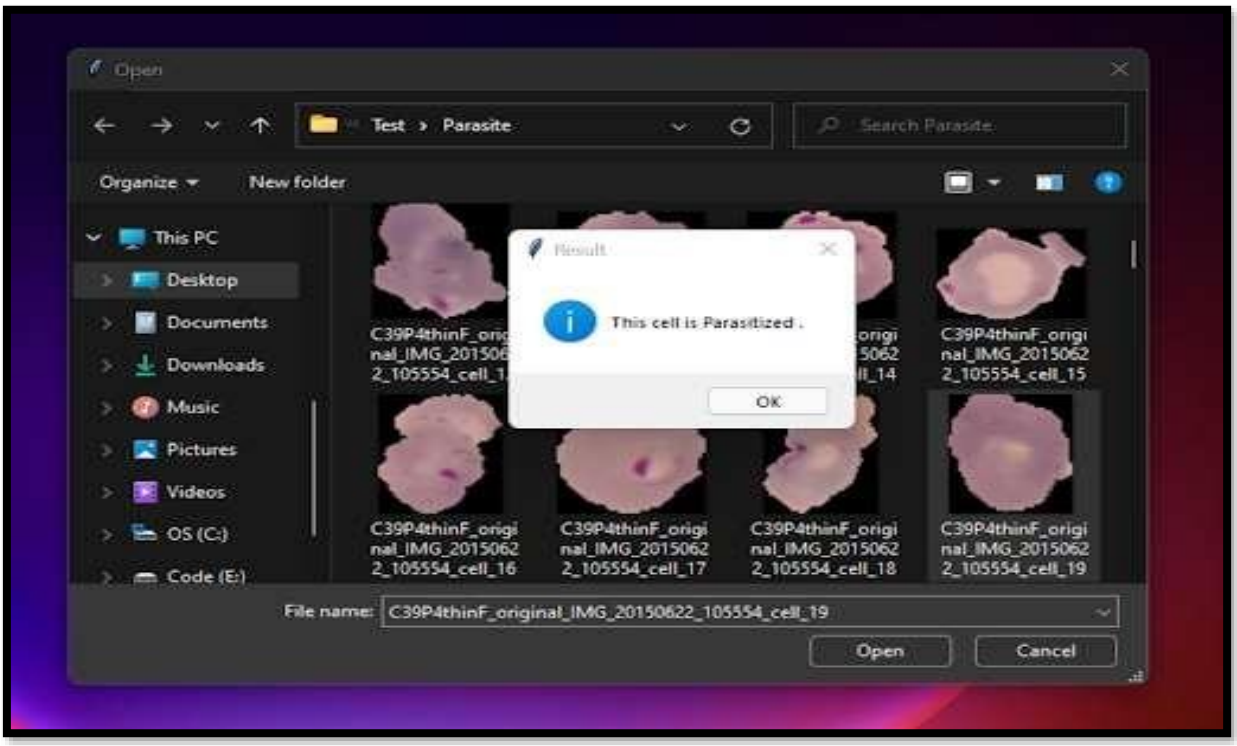

Figure 6.3: Output- Parasitized cell

As we can see in the above figure, our model correctly predicts the parasitized cells.

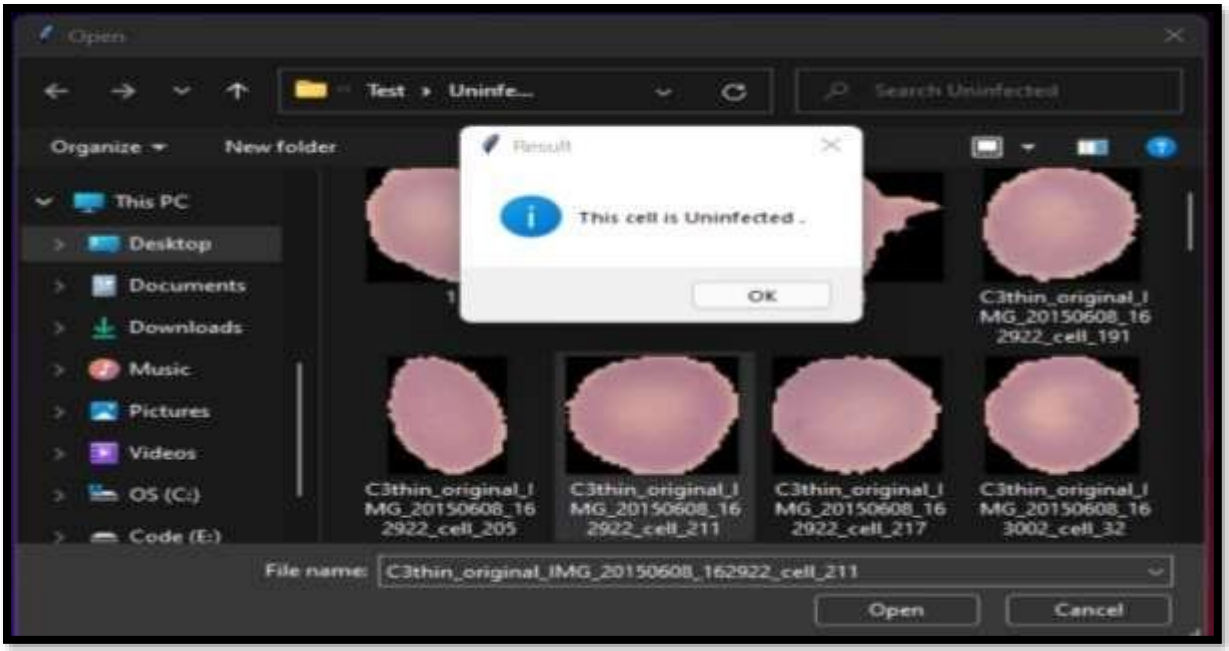

Figure 6.4: Output-Uninfected cell

As we can see in the above image, our model correctly predicts the Uninfected cells.

CHAPTER 7

CONCLUSION

We tested a variety of pre- and post-processing approaches on the malaria dataset to see how they affected the results. Pre-processing procedures such stain normalisation and standardisation had little effect on the model's performance. These findings show that for models to be trained, features do not have to be on the same scale.

On the dataset, the Data Augmentation approach performed significantly better. The data augmentation method helped to improve model performance, as shown by the results. Test Time Augmentation was another effective method for improving model performance. The malaria dataset did not respond well to stain normalisation, a pre-processing approach. This could be because the stain normalisation methods used in the study were designed for H&E stains; the samples from the National Institutes of Health malaria dataset, on the other hand, require more Haematoxylin, hence the method was not useful. We tried to keep the image's semantic relevance, which resulted in superior model performance without stain normalisation. Techniques for post-processing, such as the Ensemble model, produced positive outcomes. As a result, our best-performing model does not differ significantly from the validation sets, and it runs smoothly with a 92 percent accuracy rate.

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